Physics 1101W: Error Propagation

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Measurements that we make in the lab are not certain- a measurement depends on the fidelity of the instruments used. This **Measurement Uncertainty**, and it is the most common form of **Random Error** or **Random Uncertainty**.

Note (1):

*The words **Error** and **Uncertainty** are being pretty much used interchangeably here, and that will continue to largely be the case for this course.*

This is where you have to worry about propagating error, because when we *use* measurements from our experiment to find a **Derived Quantity**, we want to know how certain (or not) we are about that value. For a more formal treatment of addition of randomly distributed measurements see[1]. Such treatments will **NOT** be used in this class, and our treatment is far less rigorous than the above.

Definition (1): Derived Quantity

*A **Derived Quantity** is a value of a property of a system that is not being directly measured. A **Derived Quantity** is found from quantities that we* did *measure directly.*

Let's quickly go over the anatomy of a measurement. Measurements are composed of a **mean** or **Central Value** and a **Deviation**.

$$\text{Measurement} : \bar{m} + \delta m \tag{1}$$

Definition (2): Mean *or* Central Value

*The **Mean** (or the **Central Value**) of a measurement is the main number we usually think of as a measurement's output- basically it's what the instrument reads. If I were to measure a mass to be* $71g$*, that is the mean value, which basically is the average of the distribution. The mean is usually denoted with a bar, ie.* $\bar{m}$*. From a sampling of* N *datapoints, we can find the mean of the measurement distribution to be*

$$\bar{m} = \sum_{n=1}^{N} \frac{m_n}{N} = \frac{m_1}{N} + \frac{m_2}{N} + \ldots + \frac{m_N}{N} = \frac{m_1 + m_2 + \ldots + m_N}{N} \tag{2}$$

where m_n *is the* n*th measurement made.*

It is worth noting (by the conceit of measurement uncertainty) that when we measure a central value, that is not a completely accurate representation of that quantity as a whole. Due to random noise, all sorts of general errors that might occur in the measurement process, instrument calibration process, etc., we can

[1] https://en.wikipedia.org/wiki/Sum_of_normally_distributed_random_variables

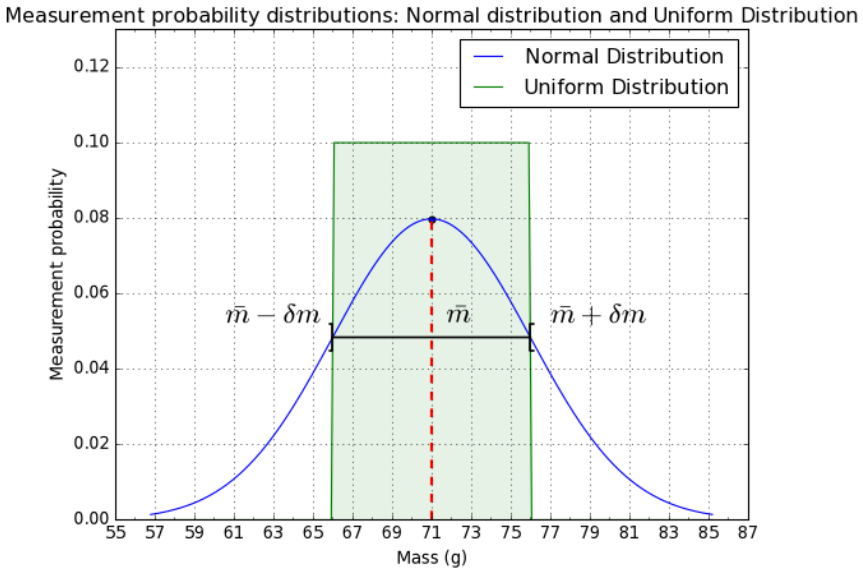


Figure 1: **Measurement Distributions** The measured mean $\bar{m}$ with respect to two types of measurement distribution. We see that the deviation δm is (half) the width of our uniform distribution, and is equal to (half) the width between the points where the probability is $\frac{1}{\sqrt{e}}$ times the maximum probability for the normal distribution.

assume that our measurements are reasonably close to the *actual* value. Mathematically, we can think of this spread as a **Normal Distribution**. At the end of the day the takeaway is the idea that the **Mean** is the most likely value for the "actual" value to be, but with decreasing likelyhood as we drift further from that center, there is a range that the "actual" value can fall within.

Definition (3): Deviation

*The **deviation** of a distribution is the range of values that are "reasonably likely" to measure around the **Mean**.*

There are a few ways to define what we call "likely"- for the uniform distribution, we can clearly see the width of the range of all possible values that can be measured, so it is natural to call this the deviation for a uniform distribution. In our measurements however, you will likely find that what you end up with is something a bit more focused around the mean value. There is a mathematical theorem known as the **Central Limit Theorem**, that states that random measurements of a certain class (including most of the types of measurements taken in a basic lab) can be represented with a **Normal Distribution**. A **Normal Distribution** (also known as a *Gaussian Distribution* or *Bell Curve*) is a distribution where the mean value is the most likely to be measured. Values away from the mean become less likely like the function e^{-x^2}.

The most commonly used measure is the **standard deviation**.

Definition (4): Average deviation and Standard Deviation

Consider a distribution $\mathcal{M}$ of N measurements $\{m_1, m_2, \ldots, m_N\}$. We define the ***Average Deviation*** *to be*

$$\text{avgdev}\left(\mathcal{M}\right) = \sum_{n=1}^{N} \frac{|m_n - \bar{m}|}{N} \tag{3}$$

Where $|\cdot|$ is the absolute value. The average deviation makes a lot of intuitive sense as a measure of how uncertain our measurement is. We sum up all of (the absolute values of) the differences *of each measurement with our* ***mean****, then divide by the number of measurements we took to get the average difference of our measurements from the mean.*
The ***standard deviation*** *is very similar to the average deviation, where the only difference is that we square the numerator of each term.*

$$\sigma\left(\mathcal{M}\right) = \sum_{n=1}^{N} \frac{\left(m_n - \bar{m}\right)^2}{N} \tag{4}$$

Materially, you can use pretty much the same intuition that we discussed for the average deviation to understand why the standard deviation is a reasonable measurement of uncertainty, since all that squaring the numerator does is cause larger differences from the mean to be counted a bit more, and smalled differences a bit less. The general behavior is the same. The main reasons to use the standard deviation are

1. *The standard deviation has a mathematically rigorous definition when propagating error, which is tied to the definition of the Normal Distribution.*

   Specifically, the standard deviation is the "distance" from your value where you are only $1/\sqrt{e} \approx 61\%$ as likely to make that measurement as you would be the mean.

2. *Pretty much all research uses it as the go-to measure of deviation*

The standard deviation and the average deviation are related by a factor of $\sqrt{\frac{2}{\pi}} \approx 0.8$, meaning that the average deviation is always going to be a *narrower* uncertainty margin.

The goal of error propagation is thus to find the error in a Derived Quantity from the error in the measurements we're using to *find* that quantity. To do so, we have to use some combination of 3 fundamental equations.

Lemma (1): Error Propagation

In the statement below, let us use capital letters for quantities with uncertainties, and lower case letters for constant/scalar values (things like fundamental constants- think of just normal numbers such as $2, 9.81$ or $\pi = 3.1415\ldots$)

$$\delta\left(c * A\right) = c * \delta A \tag{5}$$

$$\delta\left(A + B\right) = \delta A + \delta B \tag{6}$$

$$\delta\left(A * B\right) = \left(\bar{A} * \bar{B}\right)\left(\frac{\delta A}{\bar{A}} + \frac{\delta B}{\bar{B}}\right) \tag{7}$$

Note (2):

We can use the same equation for $A - B$ as for $A + B$, and similarly the same equation for $\frac{A}{B}$ as for $A * B$.

Note also that these equations can be combined/chained. By this, I mean that an uncertainty that we found by applying one of the above quantities can be used again in further error/uncertainty propagation equations.

Let's use an example

Example (1): Consider two masses m_1 and m_2. We can measure each of these masses on our balance. Take m_1 to be the full mass hanger that we use in labs (hook etc. included), and m_2 to be one of the $20g$ weight disks.
Say we hang m_2 on the m_1 hanger, and weigh them both together on a balance, which reads 71g. Call this mass of the combined hanger and disk m_s.
If we take for given that the uncertainty we get from massing a sample on our balance is $\pm$5g, what should you provide in your lab report as the weight (W_s) of these two masses together when hung from a spring during an experiment?

Answer:
We can find the uncertainty in W_s using Equation (5).

First finding our derived quantity's central value

$$\begin{aligned} W_s = (m_s)\text{g} &= 71\text{g} * 9.81\frac{\text{m}}{\text{s}^2} = 0.071\text{kg} * 9.81\frac{\text{m}}{\text{s}^2} \\ &= 0.69651\frac{kg * m}{s^2} = 0.70\text{N}(\text{up to sig figs}) \end{aligned}$$

Applying equation Equation (5), where m_s is our measured quantity (with uncertainty) and g is our scalar, and plugging in $m_s \to A, \quad g \to c$

$$\delta W_s = g * \delta m_s \tag{8}$$

$$\begin{aligned} \delta W_s &= 9.81\frac{\text{m}}{\text{s}^2} * 5\text{g} = 9.81\frac{\text{m}}{\text{s}^2} * 0.005\text{kg} \\ &= 0.04905\frac{\text{kg} * \text{m}}{\text{s}^2} = 0.049\text{N} \end{aligned} \tag{9}$$

Thus we would report a full measurement of our weight W_s as $W_s = 0.70 \pm 0.049\text{N}$.

One thing to be aware of here is that we are somewhat abusing notation: we used W_s above sometimes to refer to the Central Value of the measurement, and other times to refer to the measurement as a whole, including uncertainty. This is another bit of terminology you will have to get used to somewhat, unless you want to introduce even more subscripts! As a general rule, if you have doubts report quantities *with* uncertainties.

Example (2): Now consider if we were to measure each mass separately: m_1 being measured by the balance as 51g and m_2 at 24g

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Answer:
Measuring both of these quantities separately means that both m_1 and m_2 will carry uncertainty.
The total mass hanging from the spring can be expressed as

$$m_s = m_1 + m_2 \tag{10}$$

First we can find the central value of m_s just by normal addition

$$m_s = 51\text{g} + 24\text{g} = 75\text{g}$$

Which gives $W_s = 0.075\text{kg} * 9.81\frac{\text{m}}{\text{s}^2} = 0.73\text{N}$
We can find the uncertainty in m_s using Equation (6). Consider substituting in $m_1 \to A$ and $m_2 \to B$.

$$\begin{aligned} \delta(m_s) &= \delta(m_1 + m_2) = \delta m_1 + \delta m_2 \\ &= 5\text{g} + 5\text{g} \\ &= 10\text{g} \end{aligned} \tag{11}$$

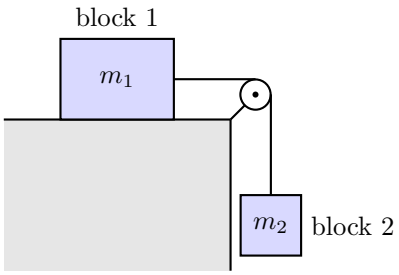
Now to find the uncertainty in weight, we want $\delta(W_s) = \delta(m_s g)$
Plug this back into Equation (5) like we did in Equation (8) to find

$$\begin{aligned} \delta(W_s) = \delta(m_s g) &= g * \delta m_s \\ &= 9.81\frac{\text{m}}{\text{s}^2} * 10\text{g} = 9.81\frac{\text{m}}{\text{s}^2} * 0.01\text{kg} \\ &= 0.091\text{N} \end{aligned} \tag{12}$$

And thus we would report $W_s = 0.73 \pm 0.091\text{g}$
It is somewhat satisfying to note that 0.73N is in fact within both of our uncertainties of the Central Value we found in Example 1! This must be the case though, since we propagated our errors correctly, the fact that Central Value we found at the start of this example is still within the uncertainty we had when measuring both masses together in Example 1 assures us that this is a possible outcome of measuring the two masses separately after measuring them together. Thus these calculations should be consistent.

As was said before, we will use many of these equations in sequence. Finally, let's practice a bit more complicated situation.

Example (3): Consider a block sliding on a horizontal surface without friction, being pulled by a hanging block on a string. The hanging block is hanging vertically off the surface, with the string connecting them over a frictionless pulley.
Find the acceleration experienced by Block 1.



Answer:
From a simple application of Newton's second law, we know that the acceleration of Block 1 should be (with the positive x axis pointing to the right

$$a_1 = g * \frac{m_2}{m_1 + m_2} \tag{13}$$

From this we can see that we will need to apply all three of the Error Propagation equations to find the uncertainty in the acceleration.
Let's work from the inside out, and as a good exercise let's make sure *not* to plug in any numbers until the end- it makes keeping track of sig figs, approximations, and just page clutter easier overall.

$$\delta(m_1 + m_2) = \delta m_1 + \delta m_2$$

From here we can note that this uncertainty can be plugged into the uncertainty for multiplication and division (Equation (7)) with $m_2 \to A, (m_1 + m_2) \to B$.

$$\begin{aligned} \delta\left(\frac{m_2}{m_1 + m_2}\right) &= \frac{m_2}{m_1 + m_2}\left(\frac{\delta m_2}{m_2} + \frac{\delta\left(m_1 + m_2\right)}{m_1 + m_2}\right) \\ &= \frac{m_2}{m_1 + m_2}\left(\frac{\delta m_2}{m_2} + \frac{\delta m_1 + \delta m_2}{m_1 + m_2}\right) \end{aligned} \tag{14}$$

Finally, we can use Equation (5) to tell us that

$$\delta\left(g\left(\frac{m_2}{m_1 + m_2}\right)\right) = g * \delta\left(\frac{m_2}{m_1 + m_2}\right) \tag{15}$$

Using Equation (14) gives us

$$\delta a_1 = \delta\left(g\frac{m_2}{m_1 + m_2}\right) = g\left(\frac{m_2}{m_1 + m_2}\right)\left(\frac{\delta m_2}{m_2} + \frac{\delta m_1 + \delta m_2}{m_1 + m_2}\right) \tag{16}$$

Example (4): Answer cont. For simplicity, take the values of m_1 and m_2 that we had before, with 5g uncertainty. First, finding the Central Value (Equation (13)):

$$a_1 = g\frac{m_2}{m_1 + m_2} \tag{17}$$

$$= 9.81\frac{\text{m}}{\text{s}^2} * \frac{24\text{g}}{51\text{g} + 24\text{g}} \tag{18}$$

The units of mass cancel and we get

$$= 9.81\frac{\text{m}}{\text{s}^2} * 0.32 \tag{19}$$

$$= 3.1\frac{\text{m}}{\text{s}^2} \text{ (up to sig figs)} \tag{20}$$

Now for uncertainty. Using Equation (16)

$$\delta a_1 = 9.81\frac{\text{m}}{\text{s}^2}\left(\frac{24\text{g}}{51\text{g} + 24\text{g}}\right)\left(\frac{5\text{g}}{24\text{g}} + \frac{5\text{g} + 5\text{g}}{51\text{g} + 24\text{g}}\right) \tag{21}$$

Again, units of mass cancel

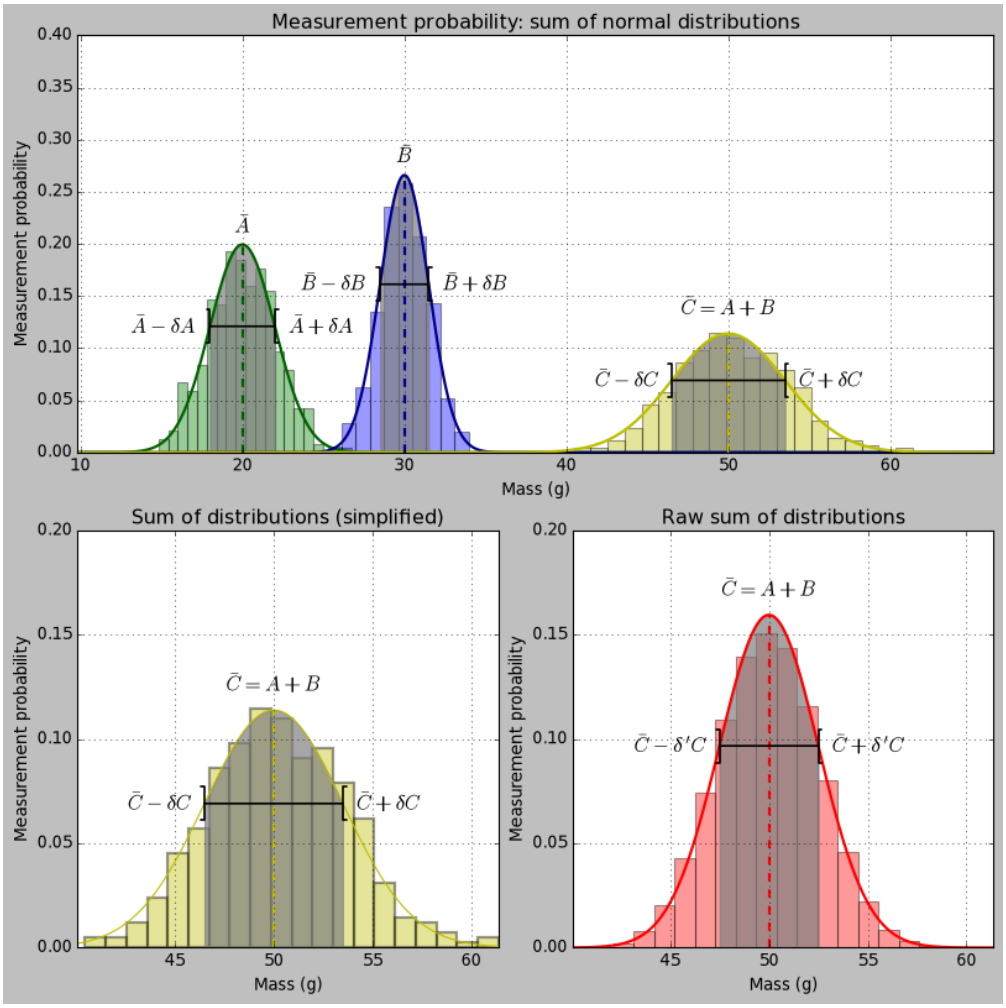
$$= 9.81\frac{\text{m}}{\text{s}^2} * \frac{24}{75}\left(\frac{5}{24} + \frac{10}{75}\right) \tag{22}$$

$$\approx 1.1\frac{\text{m}}{\text{s}^2} \tag{23}$$

Thus we would report $a_1 = 3.1 \pm 1.1\frac{\text{m}}{\text{s}^2}$

That's a pretty large uncertainty! As you can see, uncertainties can compound to become quite important, even if they seem somewhat inconsequential at first.

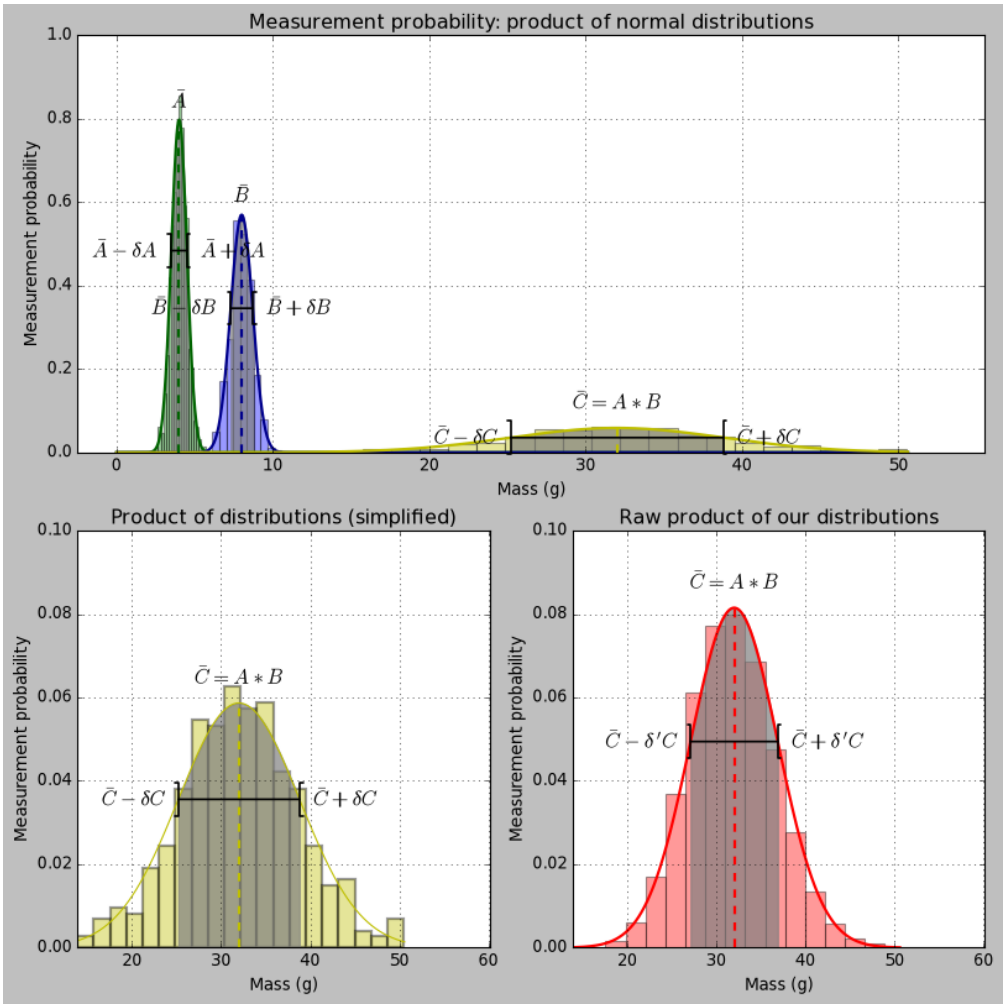
Here are some visual representations of what it looks like to add and multiply probability distributions. In both cases, I show the two starting distributions and the outcome on the top plot, as we would expect following our equations (5)(6) (7). In the bottom half, I show (left) a close up of the resulting distribution using our error propagation formulae. On the right we have what actually happens if I take distributions A and B, and combine all possible combinations of pairs of measurements from each of the distributions. For example, this would be mean that if I were adding a distribution $P = \{p_1, p_2\}$ with $Q = \{q_1, q_2\}$ then I would get a resulting distribution $R = P + Q = \{p_1 + q_1, p_1 + q_2, p_2 + q_1, p_2 + q_2\}$.



The rigorous equation for propagating error when adding is

$$\delta'(A + B) = \delta' C = \sqrt{\delta A^2 + \delta B^2} \tag{24}$$

Where δA etc. are really the standard deviations σ_A etc. here. δ stands for a typical deviation of some kind, but since these are normal distributions, I am using the standard deviation. I am also just calling it δ' to distinguish from the δ used earlier as found using our simpler propagation equations.



For multiplying

$$\delta'(A * B) = \delta' C = \left(\bar{A}^2 + (\delta A)^2\right)\left(\bar{B}^2 + (\delta B)^2\right) - \bar{A}^2\bar{B}^2 \tag{25}$$